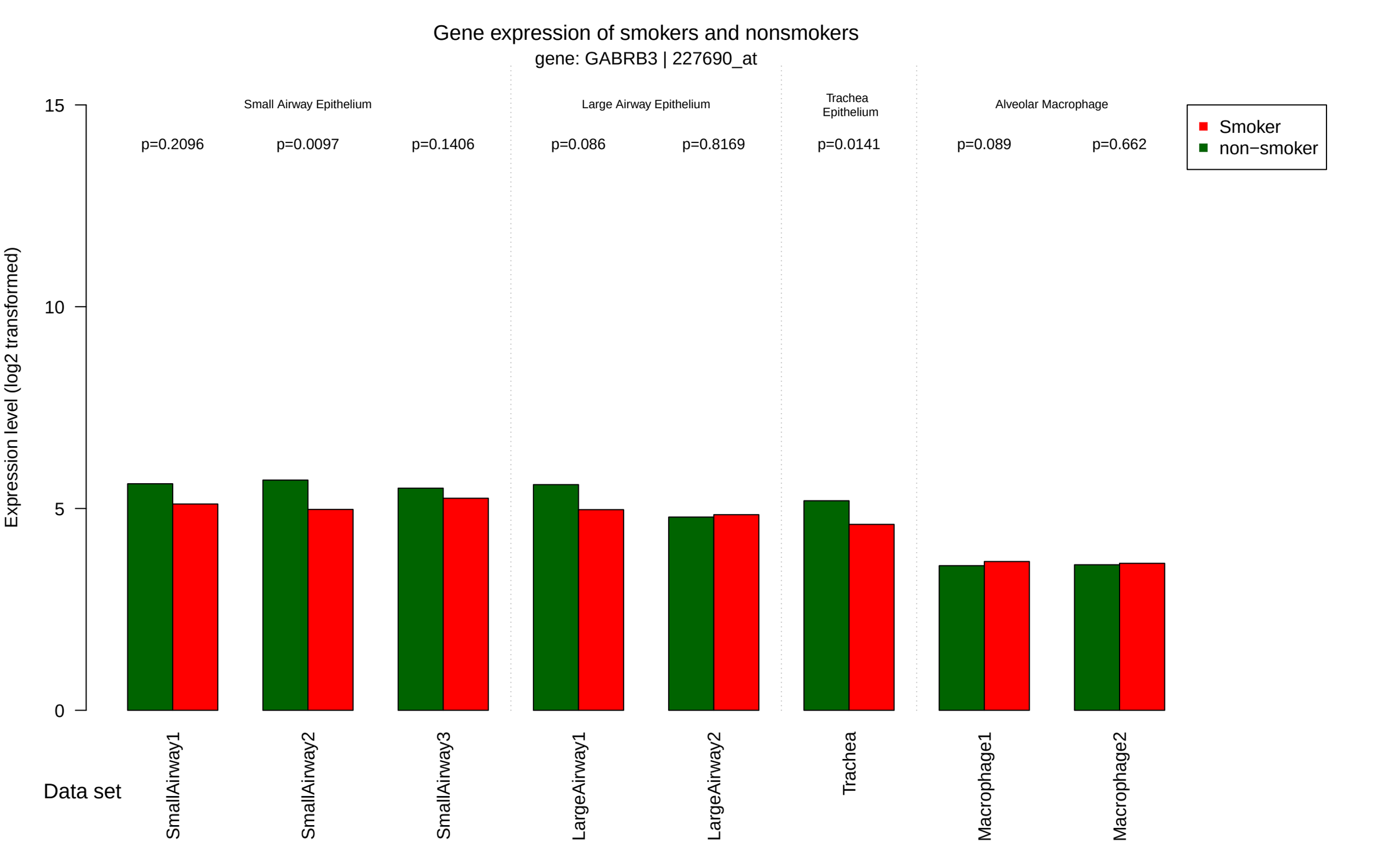
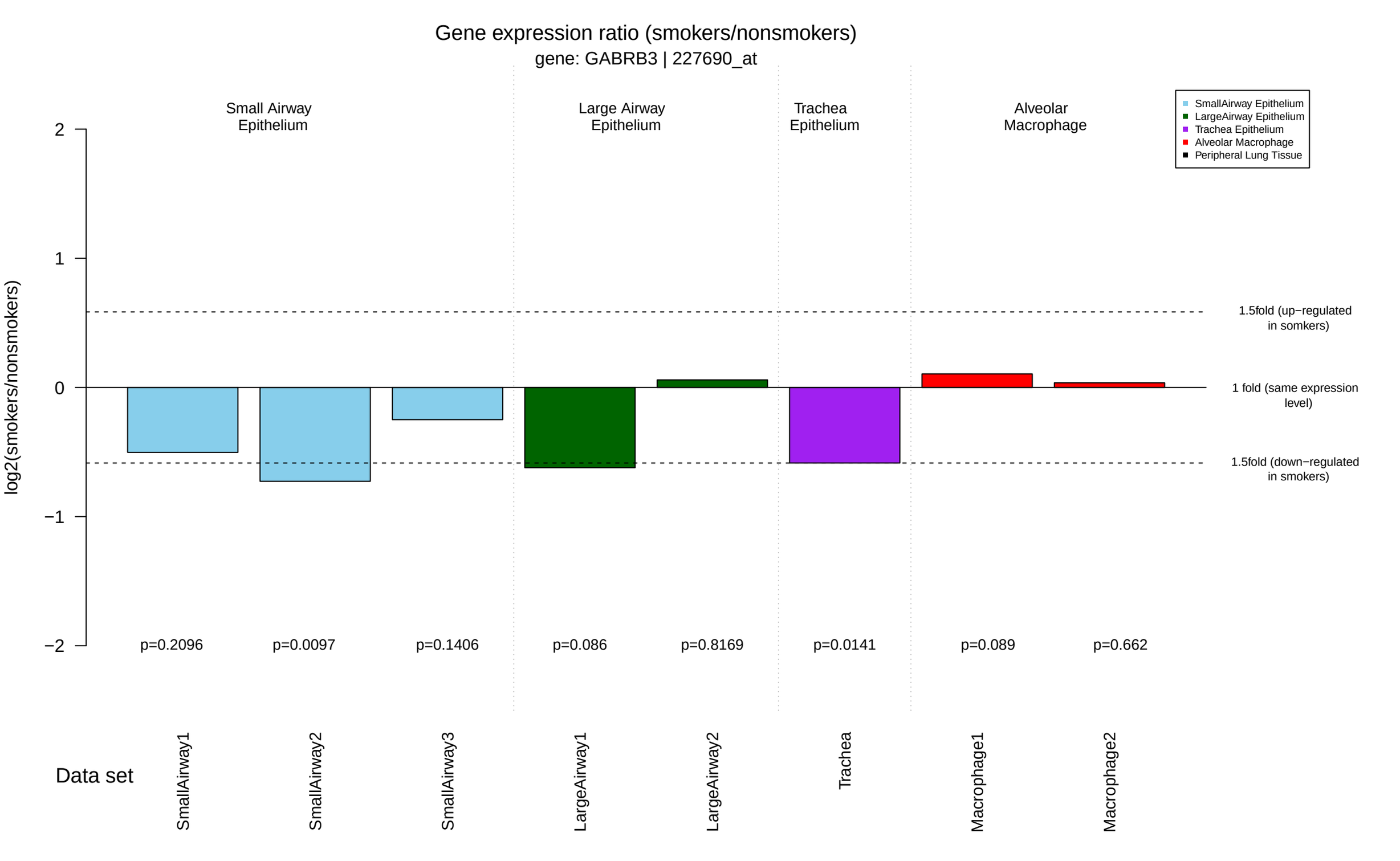
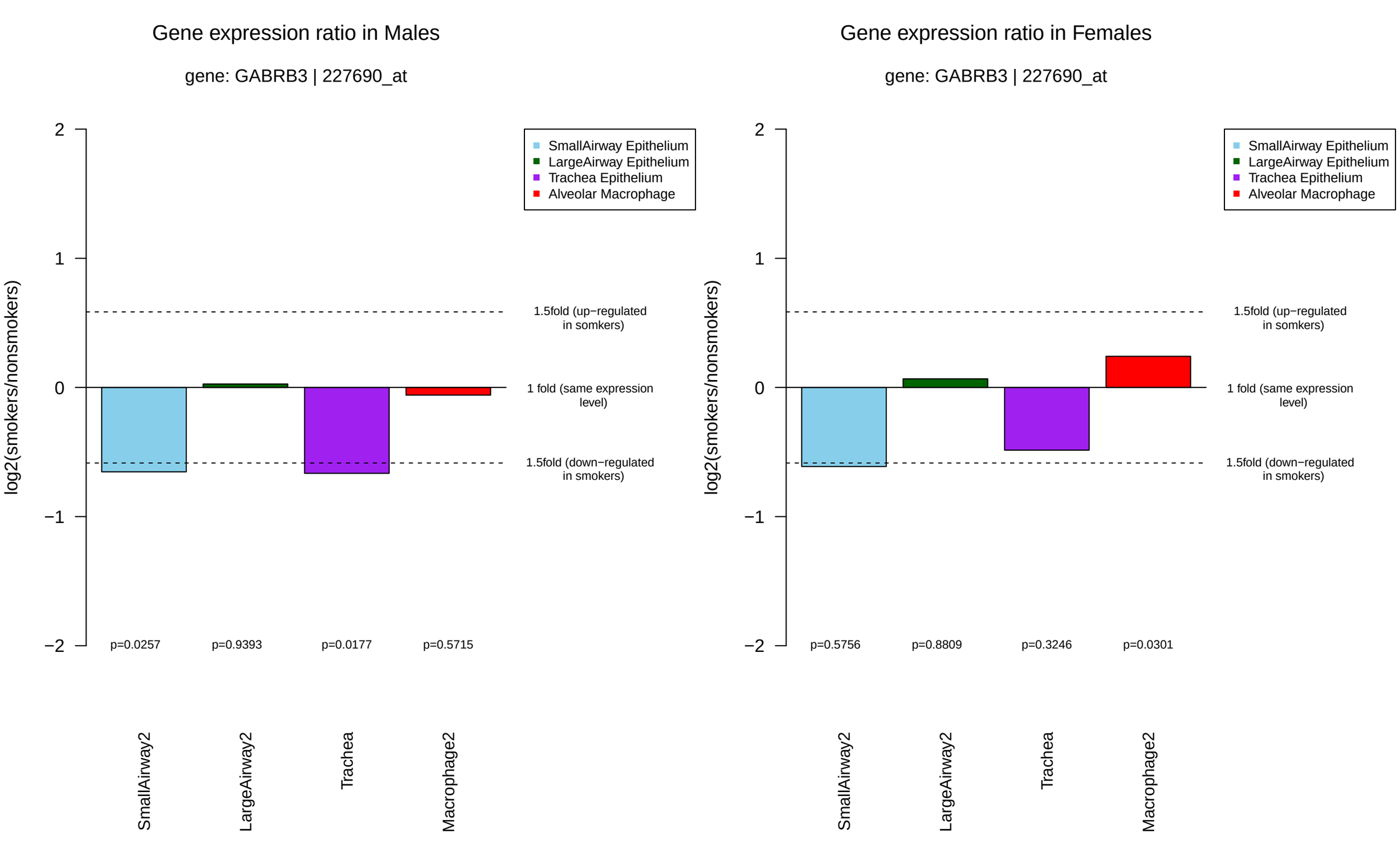
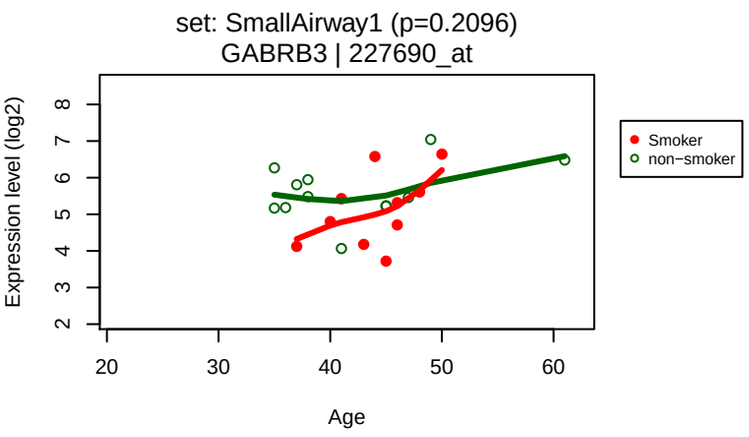
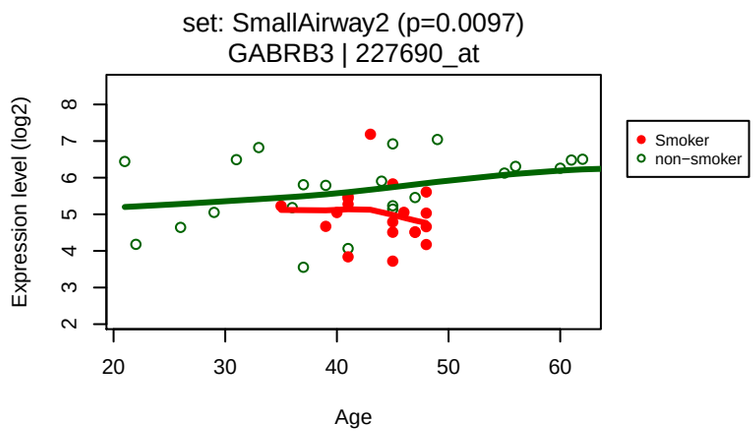
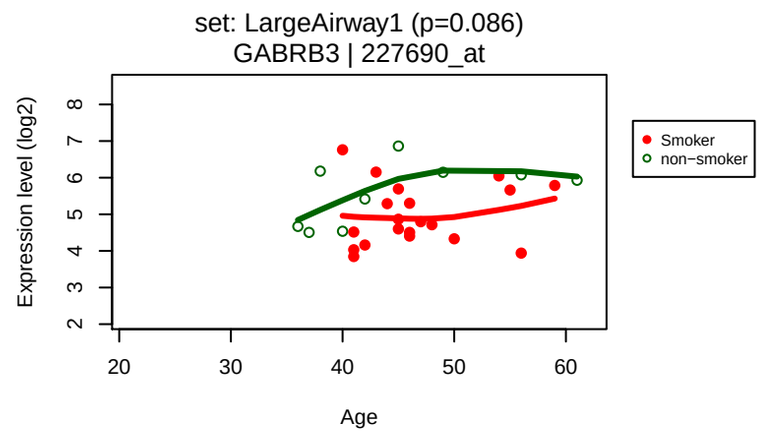

Gene expression of smokers and nonsmokers
gene: GABRB3 | 227690_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2096
p=0.0097
p=0.1406
p=0.086
p=0.8169
p=0.0141
p=0.089
p=0.662
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

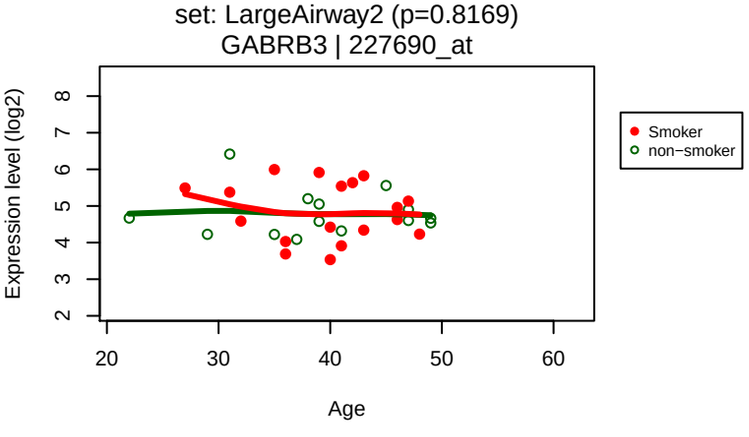

Gene expression ratio (smokers/nonsmokers)
gene: GABRB3 | 227690_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2096
p=0.0097
p=0.1406
p=0.086
p=0.8169
p=0.0141
p=0.089
p=0.662
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

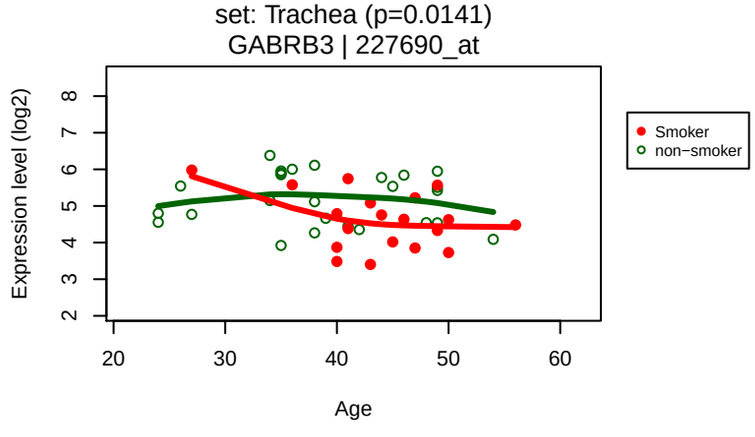

Gene expression ratio in Males
gene: GABRB3 | 227690_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0257
p=0.9393
p=0.0177
p=0.5715
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GABRB3 | 227690_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5756
p=0.8809
p=0.3246
p=0.0301
SmallAirway2
LargeAirway2
Trachea
Macrophage2

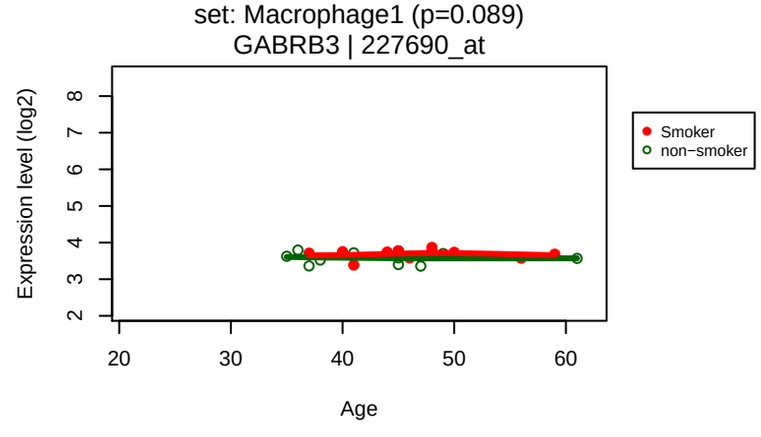

set: SmallAirway1 (p=0.2096)
GABRB3 | 227690_at
Expression level (log2)
Age
Smoker
non−smoker

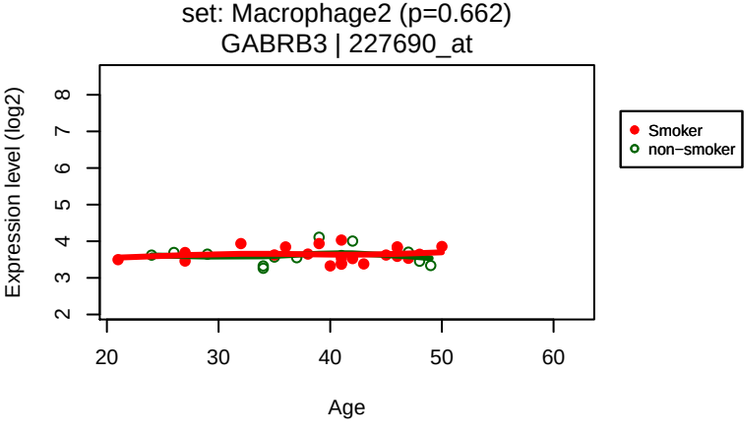

set: SmallAirway2 (p=0.0097)
GABRB3 | 227690_at
Expression level (log2)
Age
Smoker
non−smoker

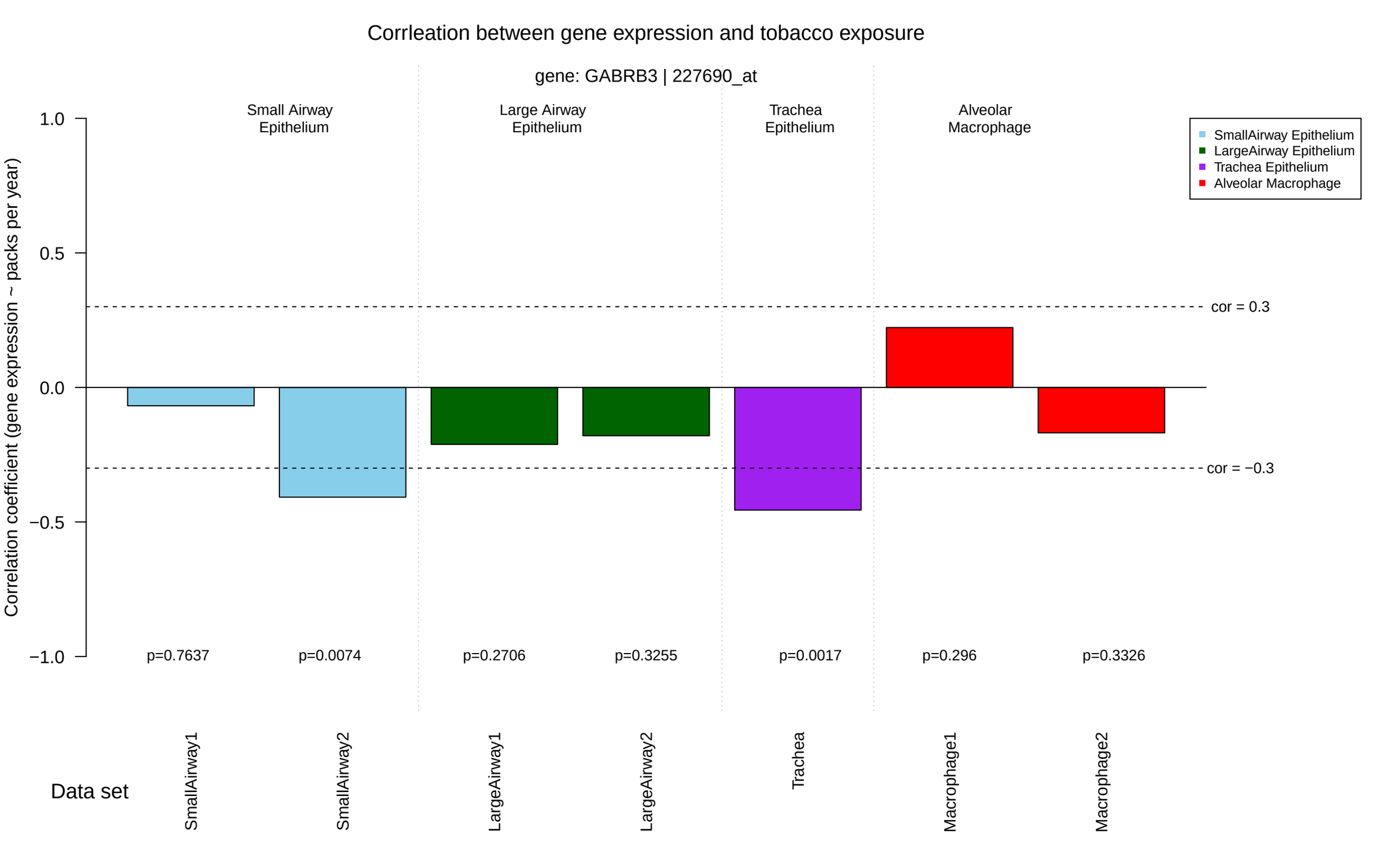

set: LargeAirway1 (p=0.086)
GABRB3 | 227690_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8169)
GABRB3 | 227690_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0141)
GABRB3 | 227690_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.089)
GABRB3 | 227690_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.662)
GABRB3 | 227690_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GABRB3 | 227690_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7637
p=0.0074
p=0.2706
p=0.3255
p=0.0017
p=0.296
p=0.3326
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2